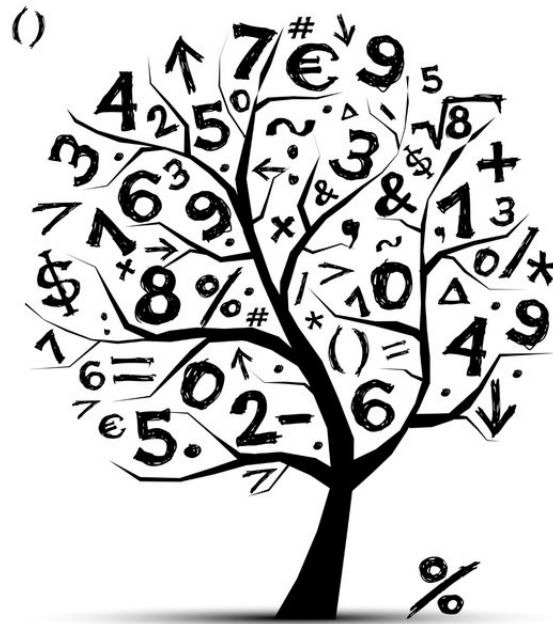


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Exercises: Factorization of matrices



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Exercise 1

Compute the LU factorization of $A = \begin{pmatrix} 4 & 3 \\ 6 & 3 \end{pmatrix}$ and verify that $LU = A$.

The first pivot is $a_{11} = 4$ and the multiplier is $\ell_{21} = \frac{6}{4} = \frac{3}{2}$:

$$\begin{pmatrix} 4 & 3 \\ 6 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 3 \\ 0 & -\frac{3}{2} \end{pmatrix} \quad (R_2 \leftarrow R_2 - \frac{3}{2}R_1)$$

Hence

$$L = \begin{pmatrix} 1 & 0 \\ \frac{3}{2} & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 4 & 3 \\ 0 & -\frac{3}{2} \end{pmatrix}, \quad LU = \begin{pmatrix} 4 & 3 \\ \frac{3}{2} \cdot 4 & \frac{3}{2} \cdot 3 - \frac{3}{2} \end{pmatrix} = \begin{pmatrix} 4 & 3 \\ 6 & 3 \end{pmatrix} = A.$$

Exercise 2

Compute the LU factorization of

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 8 & 13 \end{pmatrix}$$

and verify that $LU = A$.

The multipliers are $\ell_{21} = \frac{2}{1} = 2$, $\ell_{31} = \frac{3}{1} = 3$ and, after the first step, $\ell_{32} = \frac{2}{1} = 2$:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 8 & 13 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 3 & 8 & 13 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 2 & 4 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 3R_1)$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 2R_2)$$

Hence

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Verification:

$$\mathbf{LU} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4+1 & 6+1 \\ 3 & 6+2 & 9+2+2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 8 & 13 \end{pmatrix} = \mathbf{A}.$$

Exercise 3

Consider

$$\mathbf{A} = \begin{pmatrix} 2 & 4 & -2 \\ 1 & -1 & 5 \\ -1 & 7 & 1 \end{pmatrix}.$$

1. Check, using the leading principal minors, that $\mathbf{A}$ admits an LU factorization without row swaps.
2. Compute the LU factorization and verify that the pivots are $u_{kk} = \det(\mathbf{A}_{[k]}) / \det(\mathbf{A}_{[k-1]})$.

1. $\det(\mathbf{A}_{[1]}) = 2$, $\det(\mathbf{A}_{[2]}) = 2 \cdot (-1) - 4 \cdot 1 = -6$ and, by Laplace expansion along the first row,

$$\det(\mathbf{A}_{[3]}) = \det(\mathbf{A}) = 2 \cdot (-1 - 35) - 4 \cdot (1 + 5) + (-2) \cdot (7 - 1) = -72 - 24 - 12 = -108.$$

All the leading principal minors are nonzero, so the LU factorization exists.

2. The multipliers are $\ell_{21} = \frac{1}{2}$, $\ell_{31} = \frac{-1}{2} = -\frac{1}{2}$ and, after the first step, $\ell_{32} = \frac{9}{-3} = -3$:

$$\begin{pmatrix} 2 & 4 & -2 \\ 1 & -1 & 5 \\ -1 & 7 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 4 & -2 \\ 0 & -3 & 6 \\ -1 & 7 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - \frac{1}{2}R_1)$$

$$\begin{pmatrix} 2 & 4 & -2 \\ 0 & -3 & 6 \\ 0 & 9 & 0 \end{pmatrix} \quad (R_3 \leftarrow R_3 + \frac{1}{2}R_1)$$

$$\begin{pmatrix} 2 & 4 & -2 \\ 0 & -3 & 6 \\ 0 & 0 & 18 \end{pmatrix} \quad (R_3 \leftarrow R_3 + 3R_2)$$

Hence

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ -\frac{1}{2} & -3 & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 2 & 4 & -2 \\ 0 & -3 & 6 \\ 0 & 0 & 18 \end{pmatrix}.$$

Indeed $u_{11} = 2$, $u_{22} = \frac{-6}{2} = -3$ and $u_{33} = \frac{-108}{-6} = 18$. Verification:

$$\mathbf{LU} = \begin{pmatrix} 2 & 4 & -2 \\ 1 & 2-3 & -1+6 \\ -1 & -2+9 & 1-18+18 \end{pmatrix} = \begin{pmatrix} 2 & 4 & -2 \\ 1 & -1 & 5 \\ -1 & 7 & 1 \end{pmatrix} = \mathbf{A}.$$

Exercise 4

Consider

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Show that $\mathbf{A}$ is nonsingular but does not admit an LU factorization without row swaps. Then compute a PLU factorization $\mathbf{PA} = \mathbf{LU}$ and verify it.

We have $\det(\mathbf{A}) = 1 \cdot (4 - 1) - 2 \cdot (2 - 0) + 0 = -1 \neq 0$, but $\det(\mathbf{A}_{[2]}) = 1 \cdot 4 - 2 \cdot 2 = 0$: the LU factorization without row swaps does not exist.

Gaussian elimination: $\ell_{21} = 2$ and $\ell_{31} = 0$ (the entry a_{31} is already zero). Then the entry in position (2,2) is zero and we swap rows 2 and 3, swapping also the multipliers (now $\ell_{21} = 0$, $\ell_{31} = 2$):

$$\begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \quad (R_2 \leftrightarrow R_3)$$

The entry below the second pivot is already zero, so $\ell_{32} = 0$. Hence

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Verification:

$$\mathbf{PA} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 4 & 1 \end{pmatrix}, \quad \mathbf{LU} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 4 & 0+1 \end{pmatrix} = \mathbf{PA}.$$

Exercise 5

Compute a PLU factorization of

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 1 \\ 2 & 1 & 3 \\ 4 & 5 & 4 \end{pmatrix},$$

verify that $\mathbf{PA} = \mathbf{LU}$, and compute $\det(\mathbf{A})$ from $\mathbf{U}$.

Since $a_{11} = 0$, we swap rows 1 and 2 (no multiplier has been computed yet). Then $\ell_{21} = \frac{0}{2} = 0$, $\ell_{31} = \frac{4}{2} = 2$ and, after the first step, $\ell_{32} = \frac{3}{1} = 3$:

$$\begin{pmatrix} 0 & 1 & 1 \\ 2 & 1 & 3 \\ 4 & 5 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 & 3 \\ 0 & 1 & 1 \\ 4 & 5 & 4 \end{pmatrix} \quad (R_1 \leftrightarrow R_2)$$

$$\begin{pmatrix} 2 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 3 & -2 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 2R_1)$$

$$\begin{pmatrix} 2 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -5 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 3R_2)$$

Hence

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -5 \end{pmatrix}.$$

Verification:

$$\mathbf{PA} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 1 & 1 \\ 4 & 5 & 4 \end{pmatrix}, \quad \mathbf{LU} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 1 & 1 \\ 4 & 2+3 & 6+3-5 \end{pmatrix} = \mathbf{PA}.$$

One row swap was used ($s = 1$), hence $\det(\mathbf{A}) = -(2 \cdot 1 \cdot (-5)) = 10$.

Exercise 6

Compute the LU factorization of

$$\mathbf{A} = \begin{pmatrix} 3 & 1 & 2 \\ 6 & 3 & 4 \\ 3 & 1 & 5 \end{pmatrix}$$

and use it to compute $\det(\mathbf{A})$ and $\det(\mathbf{A}^{-1})$.

The multipliers are $\ell_{21} = \frac{6}{3} = 2$ and $\ell_{31} = \frac{3}{3} = 1$:

$$\begin{pmatrix} 3 & 1 & 2 \\ 6 & 3 & 4 \\ 3 & 1 & 5 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 1 & 2 \\ 0 & 1 & 0 \\ 3 & 1 & 5 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 3 & 1 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_1)$$

The entry in position (3,2) is already zero, so $\ell_{32} = 0$ and

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 3 & 1 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

No row swaps were used, hence

$$\det(\mathbf{A}) = \det(\mathbf{L}) \det(\mathbf{U}) = 1 \cdot (3 \cdot 1 \cdot 3) = 9, \quad \det(\mathbf{A}^{-1}) = \frac{1}{9}.$$

Exercise 7

Consider, for $k \in \mathbb{R}$, the matrix

$$\mathbf{A}(k) = \begin{pmatrix} 1 & 1 & 0 \\ 1 & k & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

1. For which values of k is $\mathbf{A}(k)$ nonsingular and admits an LU factorization without row swaps? Compute it.
2. For $k = 1$, compute a PLU factorization.

1. The leading principal minors are

$$\det(\mathbf{A}_{[1]}) = 1, \quad \det(\mathbf{A}_{[2]}) = k - 1, \quad \det(\mathbf{A}_{[3]}) = 1 \cdot (k - 1) - 1 \cdot (1 - 0) + 0 = k - 2.$$

They are all nonzero if and only if $k \neq 1$ and $k \neq 2$. For these values, $\ell_{21} = 1$,

$\ell_{31} = 0$ and $\ell_{32} = \frac{1}{k-1}$:

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & k-1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - R_1)$$

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & k-1 & 1 \\ 0 & 0 & \frac{k-2}{k-1} \end{pmatrix} \quad (R_3 \leftarrow R_3 - \frac{1}{k-1} R_2)$$

since $1 - \frac{1}{k-1} = \frac{k-2}{k-1}$. Hence

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & \frac{1}{k-1} & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & k-1 & 1 \\ 0 & 0 & \frac{k-2}{k-1} \end{pmatrix},$$

and indeed $u_{11}u_{22}u_{33} = k-2 = \det(\mathbf{A}(k))$. For $k=2$ the matrix is singular.

2. For $k=1$ we have $\det(\mathbf{A}(1)) = -1 \neq 0$ but $\det(\mathbf{A}_{[2]}) = 0$. After $R_2 \leftarrow R_2 - R_1$ ($\ell_{21} = 1, \ell_{31} = 0$) the entry in position (2,2) is zero, so we swap rows 2 and 3 together with their multipliers (now $\ell_{21} = 0, \ell_{31} = 1$):

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - R_1)$$

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \quad (R_2 \leftrightarrow R_3)$$

Then $\ell_{32} = 0$ and

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix},$$

and

$$\mathbf{P}\mathbf{A}(1) = \mathbf{L}\mathbf{U} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$